CO5 – AT3 DAA Mock Interview Report

Course: Design and Analysis of Algorithms (DAA)

Student: Lakshmi Devi

Mock Interview Duration: Approximately 15 Minutes

1. Interview Summary

The mock interview consisted of 10 questions covering backtracking, state-space trees, pruning and bounding functions, Subset Sum, Graph Coloring, Hamiltonian Cycle, complexity classes, polynomial-time reduction, approximation algorithms, and real-world algorithm selection. The student demonstrated strong conceptual understanding and provided concise, technically accurate responses.

2. Question-wise Performance

Question	Topic	Performance
Q1	Backtracking fundamentals	Very Good
Q2	State-space tree	Very Good
Q3	Pruning, promising & bounding	Good
Q4	Subset Sum	Excellent
Q5	Graph Coloring	Excellent
Q6	Hamiltonian Cycle	Excellent
Q7	P, NP, NP-Hard, NP-Complete	Excellent
Q8	Polynomial-time reduction	Excellent
Q9	Exact vs Approximation	Very Good
Q10	Real-world algorithm selection	Excellent

3. Technical Strengths

• Strong understanding of backtracking concepts.

• Clear knowledge of state-space trees and pruning.

• Good understanding of N-Queens, Subset Sum, Graph Coloring, and Hamiltonian Cycle.

• Strong understanding of P, NP, NP-Hard, and NP-Complete complexity classes.

• Accurate explanation of polynomial-time reduction.

• Good understanding of the trade-off between optimality and execution time.

• Concise, relevant, and technically accurate communication.

4. Areas Requiring Improvement

• Provide a more detailed distinction between promising functions and bounding functions.

• Include a small example when explaining higher-level theoretical concepts.

• In scenario-based questions, compare approximation with greedy and heuristic approaches when relevant.

• Expand conceptual answers slightly to better utilize the suggested 1–1.5 minute response time.

5. Rubric Evaluation

Criterion	Level	Marks
Technical Knowledge	Level 4	23/25
Problem Solving	Level 4	23/25
Analytical Thinking	Level 4	14/15
Communication	Level 4	14/15
Confidence	Level 4	9/10
Response to Questions	Level 4	9/10
Total		**92/100**

6. Overall Performance

Excellent performance – Level 4. The student demonstrated strong conceptual knowledge of major CO5 DAA topics. Responses were direct, accurate, and well-structured. Performance was particularly strong in complexity classes, polynomial-time reduction, backtracking applications, and real-world algorithm selection.

7. Specific Suggestions for Improvement

• Use a structured format: Definition → Working → Example/Application.

• Add small examples to strengthen theoretical explanations.

• Explicitly discuss time complexity, optimality, and practical usability when comparing algorithms.

• Clearly state trade-offs in analytical and scenario-based questions.

• Maintain the same confidence and clarity during the actual interview.

Final Result: 92/100 – Excellent (Level 4)